

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Department of Electrical Engineering
2. Subject Name: Process Control and Instrumentation L-T-P: 3-1-0 Credits:4
3. Pre-requisites: NIL
4. Syllabus and reference books:

Syllabus:

Understanding process control loop and salient components, Actuators and final control element, Measurement of process variable and safety, Controller design, Industrial Automation and control

Reference Books:

1. Process Dynamics and Control - Seborg, Edgar, Mellichamp, Wiley.
2. Chemical Process Control - An Introduction to Theory and Practice, George Stephanopoulos, Prentice Hall of India.
3. Process Control Handbook - B. G. Liptak, Chilton Book Company
4. Process Control - B. W. Bequette, Prentice-Hall

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Module 1: Understanding process control loop and salient components	8
	Introduction to process control loop and automation, Elements of feedback control, Process modeling and linearization, Block-diagram, Transfer functions and dynamic responses of various systems, systems with inverse response, process reaction curve, Transducer dynamics, frequency response, time-delay systems. Tutorial: Matlab Simulation Problem Tutorial: Numerical Problem	2
2.	Module 2: Actuators and final control element Classification of actuators: self-operated, pneumatic, electro-pneumatic, hydraulic, electric motor operated and stepper motor operated actuators, valve positioner and transmitter,	8

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	centralized control, Advantage of DCS Distributed Control sub-System, local field station, presentation and monitoring device, communication options in Distributed Control. Various types of communication protocols, communication between DCS, PLC, Drive, Telemetry. Tutorial: Problem/design on Relay ladder logic / control block Term paper: Distributed control system of power plant/gas and oil plant Term paper: SCADA in power plant/gas and oil plant SCADA (Supervisory Control and Data Acquisition Systems): Channel scanning, conversion to engineering units, data processing, distributed SCADA system.	3
Total number of hours		50